6. Which of the following statement is incorrect?
   a. 12 gram of C – 12 contains Avogadro's number of atoms.
   b. One mole of oxygen gas contains Avogadro's number of molecules.
   c. One mole of hydrogen gas contains Avogadro's number of atoms.
   d. One mole of electrons stands for 6.023×10^{23} electrons.

7. The volume occupied by 1 mole of a diatomic gas at S.T.P is
   a. 11.2 litre b. 5.6 litre
   c. 22.4 litre d. 44.8 litre

8. In the nucleus of $_{20}Ca^{40}$, there are
   a. 20 protons and 40 neutrons
   b. 20 protons and 20 neutrons
   c. 20 protons and 40 electrons
   d. 40 protons and 20 electrons

9. The gram molecular mass of oxygen molecule is
   a. 16 g b. 18 g
   c. 32 g d. 17 g

10. 1 mole of any substance contains ____ molecules.
    a. 6.023×10^{23} b. 6.023×10^{-23}
    c. 3.0115×10^{23} d. 12.046×10^{23}

II. Fill in the blanks

1. Atoms of different elements having _____ mass number, but _____ atomic numbers are called isobars.
2. Atoms of one element can be transmuted into atoms of other element by _____
3. The sum of the numbers of protons and neutrons of an atom is called its _____
4. Relative atomic mass is otherwise known as _____
5. The average atomic mass of hydrogen is _____ amu.
6. If a molecule is made of similar kind of atoms, then it is called _____ atomic molecule.
7. The number of atoms present in a molecule is called its _____
8. One mole of any gas occupies _____ ml at S.T.P
9. Atomicity of phosphorous is _____

III. Match the following

1.	8 g of O_2	-	4 moles
2.	4 g of H_2	-	0.25 moles
3.	52 g of He	-	2 moles
4.	112 g of N_2	-	0.5 moles
5.	35.5 g of Cl_2	-	13 moles

IV. True or False: (If false give the correct statement)

1. Two elements sometimes can form more than one compound.
2. Noble gases are Diatomic
3. The gram atomic mass of an element has no unit
4. 1 mole of Gold and Silver contain same number of atoms
5. Molar mass of CO_2 is 42g.

V. Assertion and Reason:

Answer the following questions using the data given below:

i) A and R are correct, R explains the A.

ii) A is correct, R is wrong.

iii) A is wrong, R is correct.

iv) A and R are correct, R doesn't explains A.

1. **Assertion:** The Relative Atomic mass of aluminium is 27

   Reason: An atom of aluminium is 27 times heavier than 1/12th of the mass of the C – 12 atom.

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